D06 · Freight Management & Mode Selection

04 · International freight: ocean, air & multimodal

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

International freight moves goods across borders by ocean (FCL/LCL containers), air, or multimodal combinations, coordinated with freight forwarders, ports, customs brokers and the documentation each leg requires.

2 WHY IT MATTERS

International lead times are long and variable — ocean transits are measured in weeks, and port congestion, blank sailings or customs holds add days that no forecast predicted. **That variability, not the average transit, is what drives safety stock and expediting cost.** The 2021–2022 container crisis showed how quickly rates and reliability can move; teams that managed only the rate were exposed.

3 HOW IT IS DONE

1. **Choose FCL, LCL or air by lane economics.** Volume, value and urgency decide; LCL adds handling and variability, air buys time at a steep premium.
2. **Plan with real transit distributions.** Use observed port-to-door times, not quoted schedules, to set safety stock and order timing.
3. **Qualify forwarders on service, not only on rate.** Schedule reliability, exception handling and visibility matter more when lead times are long.
4. **Keep documentation right first time.** Commercial invoice, packing list, bill of lading and certificates of origin errors cost days at the border.

IN PRACTICE

Consolidate international volume with the fewest possible freight forwarders

Negotiating power with ocean carriers and airlines is a direct function of committed volume.

Prefer annual contracts over pure spot for the majority of volume

Ocean rates are extremely volatile. Annual contracts with review clauses provide both price and capacity certainty.

Implement real-time container tracking from the day of booking

Visibility that arrives only when the container is already late has limited value.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Manufacturing company, $4.8M ocean freight/year — Three different freight forwarders by region with no volume consolidation.

Initiative	Savings	Additional impact
Consolidation to 1 global forwarder	-14% rate	Preferred carrier access
Annual contracts vs pure spot	-8%	Capacity certainty
Real-time container tracking	—	4 potential stockouts mitigated
Freight invoice audit	$84K recovered	Process discipline

Result: Total savings $720K/year (15%). The largest single lever was consolidating volume under one freight forwarder.

HOW IT IS MEASURED

Ocean Freight Cost per TEU

Total ocean freight cost / Number of TEUs

Benchmark: Compare against the relevant CCFI corridor index

Air Freight Cost per kg

Total air freight cost / Kilograms shipped

Benchmark: Compare against Baltic Air Freight Index for the corridor

WHAT YOU WOULD USE

International Freight Platforms

Flexport — Digital freight forwarding with end-to-end visibility.

project44 / Descartes — Real-time ocean and air visibility with predictive ETAs.

4 SO WHAT

Manage international freight by its variability. Pick the mode by lane economics, plan with observed transit times, qualify forwarders on reliability and get documentation right the first time.

From the SCRA v2.2 reference architecture · mrsupplychain.ai · © 2026 Ismael López · Mr. Supply Chain® México